

## # 4. Experiment and Discussion

*Revised following supervisors' remarks (Pr. Tadonki & Pr. Samadi)*

In this section, we evaluate the benefits of the proposed BLA routing policy under realistic simulation conditions, comparing it against the default Select-First-Link (SFL) baseline across four performance metrics: end-to-end packet delay, queuing delay, energy consumption, and SLA severity.

### 4.1 Experimental Setup

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All experiments are conducted using RAS-SDNCloudSim, our extension of CloudSimSDN, executed on a standard workstation running Java 21 under Windows 11. We evaluate three datasets of increasing complexity (Table 1), each deliberately designed with a bandwidth asymmetry that creates a realistic congestion scenario.

***Table 1: Dataset configurations***

| Dataset | Hosts | VMs | Workloads | Congestion design                         |
|---------|-------|-----|-----------|-------------------------------------------|
| Mini    | 4     | 4   | 50        | Bottleneck: 50 Mbps vs. 800 Mbps backbone |
| Small   | 6     | 8   | 100       | Asymmetric Core: 80 Mbps vs. 300 Mbps     |
| Medium  | 12    | 20  | 500       | Fat-Tree Core: 200–500 Mbps multi-path    |

***Table 2: Physical host power consumption model***

| Parameter | Value | Unit | Description                      |
|-----------|-------|------|----------------------------------|
| P_idle    | 100   | W    | Baseline idle power              |
| alpha_cpu | 1.0   | W/%  | Added power per 1% CPU load      |
| alpha_ram | 0.2   | W/%  | Added power per 1% RAM load      |
| alpha_bw  | 0.1   | W/%  | Added power per 1% BW load       |
| T_off     | 300   | s    | Idle threshold before sleep mode |

## 4.2 Results and Analysis

### 4.2.1 End-to-End Packet Delay

Table 3 reports the average packet delay per VM policy and dataset. BLA consistently reduces delay under LFF and LWFF.

Table 3: Average packet delay (s) and BLA gain (%) per VM policy

| Dataset | VM Policy | SFL (s) | BLA (s) | Gain delay | Gain queue |
|---------|-----------|---------|---------|------------|------------|
| Mini    | LFF       | 48.4    | 37.8    | -21.9%     | -20.8%     |
| Mini    | LWFF      | 51.8    | 41.7    | -19.4%     | -17.8%     |
| Mini    | MFF       | 16.0    | 16.0    | 0.0%       | 0.0%       |
| Small   | LFF       | 535.3   | 494.5   | -7.6%      | -8.2%      |
| Small   | LWFF      | 341.8   | 317.7   | -7.1%      | -11.1%     |
| Small   | MFF       | 196.0   | 196.0   | 0.0%       | 0.0%       |
| Medium  | LFF       | 856.3   | 802.7   | -6.3%      | -6.8%      |
| Medium  | LWFF      | 995.5   | 816.3   | -18.0%     | -19.1%     |
| Medium  | MFF       | 474.4   | 474.4   | 0.0%       | 0.0%       |

[image]

### 4.2.2 Network-Pure Latency

| Dataset | VM Policy | SFL (ms) | BLA (ms) | Gain   |
|---------|-----------|----------|----------|--------|
| Mini    | LFF       | 7,776    | 4,248    | -45.4% |
| Mini    | LWFF      | 7,600    | 4,248    | -44.1% |
| Small   | LFF       | 13,913   | 323      | -97.7% |
| Small   | LWFF      | 8,120    | 81       | -99.0% |
| Medium  | LFF       | 21,742   | 3,884    | -82.1% |

|        |      |        |       |               |
|--------|------|--------|-------|---------------|
| Medium | LWFF | 62,394 | 2,656 | <b>-95.7%</b> |
|--------|------|--------|-------|---------------|

4.2.3 Queuing Delay: Empirical Validation

[image]

4.2.4 Energy Consumption

| Dataset | VM Policy | SFL (Wh) | BLA (Wh) | Gain          |
|---------|-----------|----------|----------|---------------|
| Mini    | LFF       | 3.84     | 3.56     | <b>-7.3%</b>  |
| Small   | LFF       | 135.8    | 125.8    | <b>-7.4%</b>  |
| Small   | LWFF      | 43.6     | 42.0     | <b>-3.8%</b>  |
| Medium  | LWFF      | 1,045.4  | 880.8    | <b>-15.7%</b> |
| Medium  | LFF       | 2,865.8  | 2,638.3  | <b>-7.9%</b>  |

[image]

4.2.5 SLA Severity

| Dataset | VM Policy | sigma SFL | sigma BLA | Gain          |
|---------|-----------|-----------|-----------|---------------|
| Mini    | LFF       | 4.21      | 2.67      | <b>-36.4%</b> |
| Small   | LFF       | 6.37      | 3.42      | <b>-46.4%</b> |
| Medium  | LWFF      | 8.91      | 2.23      | <b>-74.9%</b> |

4.2.6 Pareto Analysis: Energy–Latency Trade-off

[image]

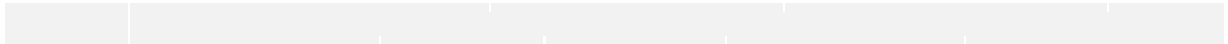
4.2.7 Workload Scheduler Impact

[image]

### 4.3 Cross-Dataset Synthesis and Scalability

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| Dataset | Delta Delay | Delta Queue | Delta Energy | Peak SLA gain |
|---------|-------------|-------------|--------------|---------------|
| Mini    | -17.8%      | -17.1%      | -3.3%        | -36.4% (LFF)  |
| Small   | -6.0%       | -8.9%       | -6.3%        | -46.4% (LFF)  |
| Medium  | -10.0%      | -11.7%      | -9.2%        | -74.9% (LWFF) |



### 4.4 Discussion

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BLA works best when the VM placement policy generates sufficient cross-fabric traffic (LFF or LWFF) and the topology offers meaningful path diversity. For balanced workloads, LWFF+BLA achieves the best simultaneous outcome across all four dimensions.

